# The Goldwasser–Sipser theorem

A public-coin proof by counting tickets

A private-coin verifier may conceal its random choices from the prover. A public-coin verifier, usually called Arthur, sends the prover its fresh random bits at each turn and makes its final decision deterministically. Revealing the coins of an existing protocol can destroy its soundness. The theorem below constructs a *different* protocol.

**Theorem (Goldwasser–Sipser [GS86]).** Suppose a language has a private-coin interactive proof with $t = t(n)$ verifier–prover message pairs, completeness at least $2/3$, and soundness at most $1/3$. Then it has a public-coin interactive proof with $t + 1$ message pairs, the same completeness and soundness bounds, and polynomial overhead.
Here $t$ may be polynomial in the input length. Thus, when messages are counted individually and the verifier speaks first,

$$\mathsf{IP}[2t] \subseteq \mathsf{AM}[2t + 2].$$

For constant $t$, the separate constant-round collapse for public-coin proofs gives the usual two-message class $\mathsf{AM}$. Removing private coins and collapsing the number of rounds are distinct results; only the first is proved here. See [Gol08, Appendix F.2] for both results.

**The proof idea.** Count how many of the original verifier's random tapes an optimal prover can make accepting. This count satisfies a sum/max recurrence. Replace each weight $w$ in the sum by $w$ numbered tickets. Hashing lets Arthur require a surviving ticket and then test its justification one level deeper in the conversation. At the end, membership is directly checkable.

The soundness argument is the following invariant:

> *With $d$ checks left, an overstatement by a factor $\Gamma^d$ cannot become an overstatement smaller than $\Gamma^{d-1}$ in one step, unless the random hash hits a small exceptional set.*

The factor $\Gamma$ will be polynomial in $t$. Parallel amplification supplies the large initial gap; its proof is included in the appendix.

## 1 Count accepting random tapes

Fix an input $x$ and a private-coin verifier $V$. Its conversation has the form

$$q_1, a_1, q_2, a_2, \ldots, q_t, a_t,$$

where $q_i$ is a verifier question and $a_i$ is a prover answer. After the final answer, $V$ accepts or rejects. Pad shorter conversations and messages to fixed polynomial bounds. We may regard all of $V$'s randomness as a tape $\rho \in \{0,1\}^m$ chosen in advance. Write

$$N = 2^m.$$

A tape is *consistent* with a partial transcript if replaying $V$ with that tape and the recorded prover answers produces exactly the recorded verifier questions.

For a partial transcript $\tau$ ending after a prover answer, let $W(\tau)$ be the maximum number of consistent tapes on which *one legal continuation strategy* makes $V$ accept. A legal strategy chooses its answers from the visible transcript, not from the hidden tape. We also allow the empty transcript $\varnothing$.

These are counts in the original space of $N$ tapes, *not conditional probabilities*. In particular,

$$\frac{W(\varnothing)}{N} = \text{the original prover's optimal acceptance probability.} \tag{1}$$

Randomized prover strategies cannot improve this optimum: averaging over the prover's own randomness leaves some deterministic strategy at least as successful.

### The recurrence

For a nonterminal transcript, the crucial identity is

$$\boxed{W(\tau) = \sum_{q} \max_{a} W(\tau, q, a).} \tag{2}$$

Here the sum and maximum range over all padded messages of the appropriate lengths. Impossible questions contribute zero. This is the counting structure underlying the Goldwasser–Sipser conversion [GS86, Section 4.2].

To prove (2), partition the tapes consistent with $\tau$ according to the next question $q$. Upon seeing $q$, the prover chooses one answer $a$ and then a continuation strategy. For this question it can win on at most $\max_a W(\tau, q, a)$ tapes. Conversely, these optimal choices can be made independently for different questions, since the prover sees which question was asked.

**The maximum is essential.** The prover must choose one answer for a given visible question. It cannot choose a different answer for each hidden tape producing that same question.

### At the leaves, membership is easy

For a complete transcript $T$, define

$$R_T = \{\rho \in \{0,1\}^m : \ V(x;\rho) \text{ follows } T \text{ and accepts}\}. \tag{3}$$

There are no future prover choices, so

$$W(T) = |R_T|. \tag{4}$$

Arthur can check $\rho \in R_T$ in polynomial time: replay $V$ on tape $\rho$, feed it the answers recorded in $T$, check *every* verifier question against $T$, and check acceptance. Verifying only the final acceptance predicate would not suffice.

## 2 Turn weights into tickets

Fix a nonterminal transcript $\tau$, and abbreviate $w_q = \max_a W(\tau, q, a)$. Give question $q$ exactly $w_q$ formal tickets:

$$(q, 1), (q, 2), \ldots, (q, w_q).$$

The resulting set is

$$S_\tau = \{(q, j) : 1 \le j \le w_q\}, \qquad \boxed{|S_\tau| = \sum_q w_q = W(\tau).} \tag{5}$$

For example, questions with weights $2, 0, 4$ contribute two, zero, and four tickets, giving six tickets altogether. The ticket numbers are merely indices; they do not encode random tapes.

Membership of a ticket has the useful interpretation

$$(q, j) \in S_\tau \iff \exists a\ W(\tau, q, a) \ge j, \qquad 1 \le j \le N. \tag{6}$$

Thus, to justify a ticket $(q, j)$, Merlin can supply an answer $a$ and make a new lower-bound claim one level deeper in the conversation.

> **Hash $(q, j)$, not $(q, a, j)$.** The answer $a$ is a witness that the ticket exists; it is not part of the ticket's identity. Including it would count all answers instead of taking the best answer, replacing the crucial maximum by a sum.

Arthur generally cannot test $(q, j) \in S_\tau$ directly. That is not required: the next round tests the new claim $W(\tau, q, a) \ge j$. Only the final membership test, for $R_T$, will be performed directly.

Every ticket has polynomial length. The question is a polynomial-length string, and $1 \le j \le N = 2^m$ can be encoded using $m + 1$ bits. Fix a unique padded binary encoding of each pair.

## 3 A random filter for large sets

The hashing ingredient is a loose-gap form of the usual set-size lower-bound test [GS86, Section 4.1]. Arthur chooses a random filter so that a large set probably has a surviving element, while a much smaller set probably has none.

**Lemma 1** (Random-filter lemma)**.** Let $0 < \eta < 1/2$ and $\Gamma = \lceil 2/\eta^2 \rceil$. Given a positive integer $K$ and a universe $\{0,1\}^u$, Arthur can publicly choose an efficiently testable random subset $H \subseteq \{0,1\}^u$ such that, for every fixed set $S \subseteq \{0,1\}^u$,

$$|S| \ge K \implies \Pr[H \cap S \ne \emptyset] \ge 1 - \eta, \tag{7}$$

$$|S| \le K/\Gamma \implies \Pr[H \cap S \ne \emptyset] \le \eta. \tag{8}$$

The description of $H$ has size polynomial in $u$ and $\log K$. Arthur need not know $S$ to choose $H$ or test membership in $H$.

*Proof.* First suppose $\eta K \geq 1$. Choose a power of two $M = 2^s$ satisfying

$$\eta K/2 < M \leq \eta K.$$

Choose a uniformly random affine map

$$h : \mathbb{F}_2^u \longrightarrow \mathbb{F}_2^s, \qquad h(z) = Az + c,$$

and take $H = \{z : h(z) = 0\}$. Arthur's public bits specify $A$ and $c$.

For each $z$, the value $h(z)$ is uniform. For distinct $z, z'$, the two values are independent: $A(z - z')$ is uniform, and the independent shift $c$ makes $h(z)$ uniform independently of this difference. Hence the indicators $\mathbf{1}[h(z) = 0]$ are pairwise independent.

Put $X = |H \cap S|$ and $\mu = \mathbb{E}X$. Then

$$\mu = \frac{|S|}{M}, \qquad \mathrm{Var}(X) = |S|\frac{1}{M}\left(1 - \frac{1}{M}\right) \leq \mu.$$

If $|S| \geq K$, then $\mu \geq 1/\eta$, so Chebyshev's inequality gives

$$\Pr[X = 0] \leq \frac{\mathrm{Var}(X)}{\mu^2} \leq \frac{1}{\mu} \leq \eta.$$

If $|S| \leq K/\Gamma$, Markov's inequality gives

$$\Pr[X \geq 1] \leq \mu \leq \frac{K}{\Gamma M} \leq \frac{2}{\eta\Gamma} \leq \eta.$$

Finally, if $\eta K < 1$, let $H$ be the entire universe. A set of size at least $K \geq 1$ is nonempty, while a set of size at most $K/\Gamma < 1$ must be empty. Both assertions follow. □

**The timing matters.** The set $S$ must be fixed before the filter is chosen. Merlin may then see the filter and choose any surviving element. The small-set bound already controls the probability that *any* such element exists.

## 4 The public-coin protocol

For $t$ original verifier–prover pairs, set

$$\eta = \frac{1}{3(t+1)}, \qquad \Gamma = 18(t+1)^2. \tag{9}$$

These satisfy Lemma 1, and $(t+1)\eta = 1/3$.

First amplify the private-coin proof, without increasing its number of messages, so that its completeness is at least $2/3$ and its soundness is at most

$$\delta = \frac{1}{2\Gamma^{t+1}}. \tag{10}$$

This preprocessing uses parallel repetition [GS86, Section 4.2]. The appendix proves that $O((t+1)\log(t+2))$ copies suffice, even against fully coordinated cheating, without adding messages.

From now on, $V, m, N, W$ refer to the *amplified* verifier. Add an unused random bit if necessary so that $m \geq 1$ and $N/2$ is an integer. Write $\mathsf{Filter}(K)$ for the random-filter construction with parameters (9) on the indicated universe.

**PUBLIC-COIN PROTOCOL**

**State.** A partial transcript $\tau$ and a positive integer $K$, interpreted as the claim $W(\tau) \geq K$.
**Initialize.** Set $\tau = \varnothing$ and $K = N/2$.
**Repeat for $t$ exchanges.**

1. Arthur sends fresh public randomness specifying $H \leftarrow \mathsf{Filter}(K)$ on the binary encodings of pairs $(q, j)$.
2. Merlin sends $(q, a, j)$. Arthur checks the message lengths, $1 \leq j \leq N$, and $(q, j) \in H$. If any check fails, reject.
3. Replace $(\tau, K)$ by $((\tau, q, a), j)$.

**Final exchange.** The transcript is now complete; call it $T$. Arthur sends fresh public randomness specifying $H \leftarrow \mathsf{Filter}(K)$ on $\{0,1\}^m$. Merlin sends a tape $\rho \in \{0,1\}^m$. Arthur accepts exactly when

$$\rho \in H \quad \text{and} \quad \rho \in R_T.$$

The second condition is checked by replaying $V$ as in (3).

Merlin chooses both the simulated question $q$ and its answer $a$, subject to the ticket test. Arthur does not reproduce the original distribution of conversations. The final tape is a *witness chosen by Merlin*, not hidden randomness used by Arthur.

**Resources.** There are exactly $t + 1$ exchanges. All messages have polynomial length; Arthur only evaluates affine hashes, checks binary integers, and replays $V$. His messages consist entirely of fresh public randomness.[1]

## 5 Completeness and soundness

### Completeness: true claims stay true

On a yes-instance, $W(\varnothing) \geq (2/3)N \geq N/2$, so the initial claim is true.

Suppose the current claim $W(\tau) \geq K$ is true. By (5), the ticket set $S_\tau$ has size at least $K$. With probability at least $1 - \eta$, the filter contains a ticket $(q, j) \in S_\tau$. By (6), Merlin can choose an answer $a$ satisfying

$$W(\tau, q, a) \geq j.$$

The new claim is therefore true as well.

At the final exchange, a true claim says $|R_T| = W(T) \geq K$. With probability at least $1 - \eta$, some valid tape survives the last filter. Thus, conditional on success so far, each exchange fails with probability at most $\eta$. The honest prover is accepted with probability at least

$$1 - (t + 1)\eta = \frac{2}{3}.$$

[1] The hash length may depend on $K$, which is public. Fixed message lengths can be obtained by sending a polynomial upper bound of fresh random bits and ignoring the unused bits.

### Soundness: a large overstatement cannot disappear

We prove a stronger statement, applicable after *any* history of the public-coin protocol. Let $d$ be the number of remaining filter challenges, including the final tape challenge. Then

$$\boxed{W(\tau) \le \frac{K}{\Gamma^d} \quad \Longrightarrow \quad \Pr[\text{eventual acceptance} \mid \text{current history}] \le d\eta.} \tag{11}$$

The probability bound holds for every future strategy of Merlin. We prove it by induction on $d$.

**The final challenge: $d = 1$.** The transcript is complete, and the hypothesis gives

$$|R_T| = W(T) \le K/\Gamma.$$

By the small-set bound (8), the probability that any valid tape survives is at most $\eta$. This proves the base case.

**One descent step: $d \ge 2$.** Set $w_q = \max_a W(\tau, q, a)$. Consider the tickets that could let Merlin escape the desired next inequality:

$$E_\tau = \{(q, j) : 1 \le j \le N,\ j \le \Gamma^{d-1} w_q\}. \tag{12}$$

This set depends on the current history, but not on the new filter. Its size satisfies

$$|E_\tau| \le \Gamma^{d-1} \sum_q w_q = \Gamma^{d-1} W(\tau) \le \frac{K}{\Gamma}. \tag{13}$$

By (8), the probability that the new filter hits $E_\tau$ is at most $\eta$.

Unless that event occurs, *every* ticket $(q, j)$ that passes Arthur's check lies outside $E_\tau$. Consequently,

$$j > \Gamma^{d-1} w_q,$$

and whatever answer $a$ Merlin supplies,

$$W(\tau, q, a) \le w_q < \frac{j}{\Gamma^{d-1}}. \tag{14}$$

The next state has claimed lower bound $j$ and $d - 1$ remaining challenges, so the induction hypothesis bounds its acceptance probability by $(d-1)\eta$. Allowing acceptance with probability 1 on the exceptional event yields an overall bound of at most

$$\eta + (d-1)\eta = d\eta.$$

This completes the induction. If Merlin has no valid response, Arthur simply rejects.

**Apply the bound at the root.** On a no-instance, amplification and (1) give

$$W(\varnothing) \le \delta N = \frac{N/2}{\Gamma^{t+1}}.$$

Thus (11) applies initially with $K = N/2$ and $d = t + 1$. Every prover is accepted with probability at most

$$(t+1)\eta = \frac{1}{3}.$$

This proves soundness.

Together with the resource bounds above, this completes the theorem. □

## A Parallel amplification without extra messages

The main proof uses error reduction to $\delta = 1/(2\Gamma^{t+1})$. The subtlety is that a cheating prover can coordinate its answers across parallel copies. Its success indicators need not be independent. The following argument handles this explicitly.

**Lemma 2** (Parallel amplification)**.** A private-coin interactive proof with completeness at least $2/3$ and soundness at most $1/3$ can have both errors reduced to any $0 < \delta < 1/3$ by running $r = O(\log(1/\delta))$ copies in parallel and taking majority. The number of messages is unchanged.

### A direct-product fact for conjunction

Run $r$ verifiers in parallel with independent random tapes, batching together their questions at each turn and their answers at the next turn. Let $v_i$ be the optimal acceptance probability of copy $i$. Then

$$\max_{P} \Pr[\text{all } r \text{ copies accept}] = \prod_{i=1}^{r} v_i. \tag{15}$$

This is an exact statement, even when $P$ chooses each answer using the entire joint transcript.

Here is a proof using the same counting recurrence as before. For a fixed joint complete transcript $(T_1, \dots, T_r)$, the accepting tuples of tapes form the Cartesian product

$$R_{T_1} \times \cdots \times R_{T_r}.$$

Hence the leaf count is the product of the individual leaf counts. This product structure propagates backward through the recurrence because, for nonnegative numbers $u_i(q_i, a_i)$,

$$\sum_{q_1,\dots,q_r} \max_{a_1,\dots,a_r} \prod_{i=1}^{r} u_i(q_i, a_i) = \prod_{i=1}^{r} \left( \sum_{q_i} \max_{a_i} u_i(q_i, a_i) \right). \tag{16}$$

For a fixed tuple of questions, the answer coordinates can be optimized separately; then the sum over question tuples factors. Backward induction therefore gives a product at the root. Dividing by the number of independent tape tuples proves (15).

More generally, for a fixed subset $J \subseteq [r]$ and any joint prover strategy on a no-instance,

$$\Pr[\text{every copy in } J \text{ accepts}] \le (1/3)^{|J|}. \tag{17}$$

To deduce this from (15), leave all $r$ conversations in place, but regard the copies outside $J$ as automatically accepting at the end. Those copies have value 1; the copies in $J$ have value at most $1/3$.

### From conjunction to majority

Fix any cheating joint prover on a no-instance. Let $I_i$ indicate acceptance in copy $i$, and let $Z = \sum_{i=1}^{r} I_i$. We do *not* assume the $I_i$ are independent. Instead, expand

$$\begin{aligned}\mathbb{E}[2^Z] &= \mathbb{E}\left[\prod_{i=1}^{r}(1+I_i)\right] \\ &= \sum_{J\subseteq[r]} \mathbb{E}\left[\prod_{i\in J} I_i\right] \\ &\le \sum_{J\subseteq[r]} (1/3)^{|J|} = \left(\frac{4}{3}\right)^r,\end{aligned}$$

where the inequality is (17). Markov's inequality now gives

$$\Pr[Z \ge r/2] \le \frac{(4/3)^r}{2^{r/2}} = \left(\frac{4}{3\sqrt{2}}\right)^r. \tag{18}$$

The base $4/(3\sqrt{2})$ is strictly smaller than 1.

For completeness, the honest prover runs the copies independently. Each copy fails with probability at most $1/3$. Applying the same calculation to the independent *failure* indicators bounds the probability that at least half the copies fail by the same expression. Taking $r$ odd avoids ties, and taking

$$r \ge \frac{\log(1/\delta)}{\log(3\sqrt{2}/4)}$$

reduces both errors to at most $\delta$. Every turn still consists of one batched verifier message or one batched prover message, so there are no extra exchanges. □

## References and further reading

The theorem and its hashing-based proof originate in [GS86]. The presentation above replaces a sum of weights by a set of numbered tickets, so no grouping of weights into size classes is needed. For a full textbook treatment of the general conversion, see [Gol08, Appendix F.2.1]. For the set-size protocol and the graph-nonisomorphism application, see [AB07, Section 8.4].

[GS86] Shafi Goldwasser and Michael Sipser. *Private coins versus public coins in interactive proof systems.* Proceedings of the 18th ACM Symposium on Theory of Computing, pp. 59–68, 1986. In particular, Sections 4.1–4.2.

[Gol08] Oded Goldreich. *Computational Complexity: A Conceptual Perspective*. Cambridge University Press, 2008. Appendix F.2: round-efficient public-coin emulation and round reduction.

[AB07] Sanjeev Arora and Boaz Barak. *Computational Complexity: A Modern Approach*, web draft, January 2007. Section 8.4, "Public Coins and AM."